Day 03

Network Configuration

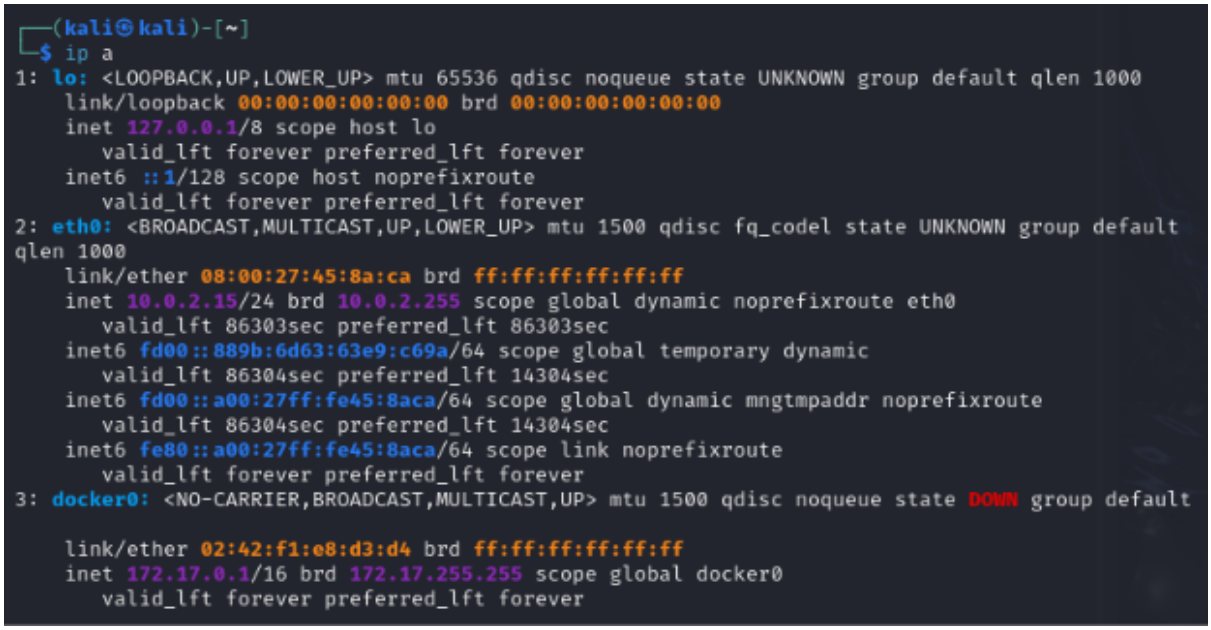


Figure 1: ip a

Displays information about the network interfaces available on the system. Helps identify the system's IP address and inspect network interface.



Figure 2: ip route

Displays the system's routing table. It shows the default gateway, network destinations, routes used to send network traffic. Helps understand how the system sends data to other networks and the internet.

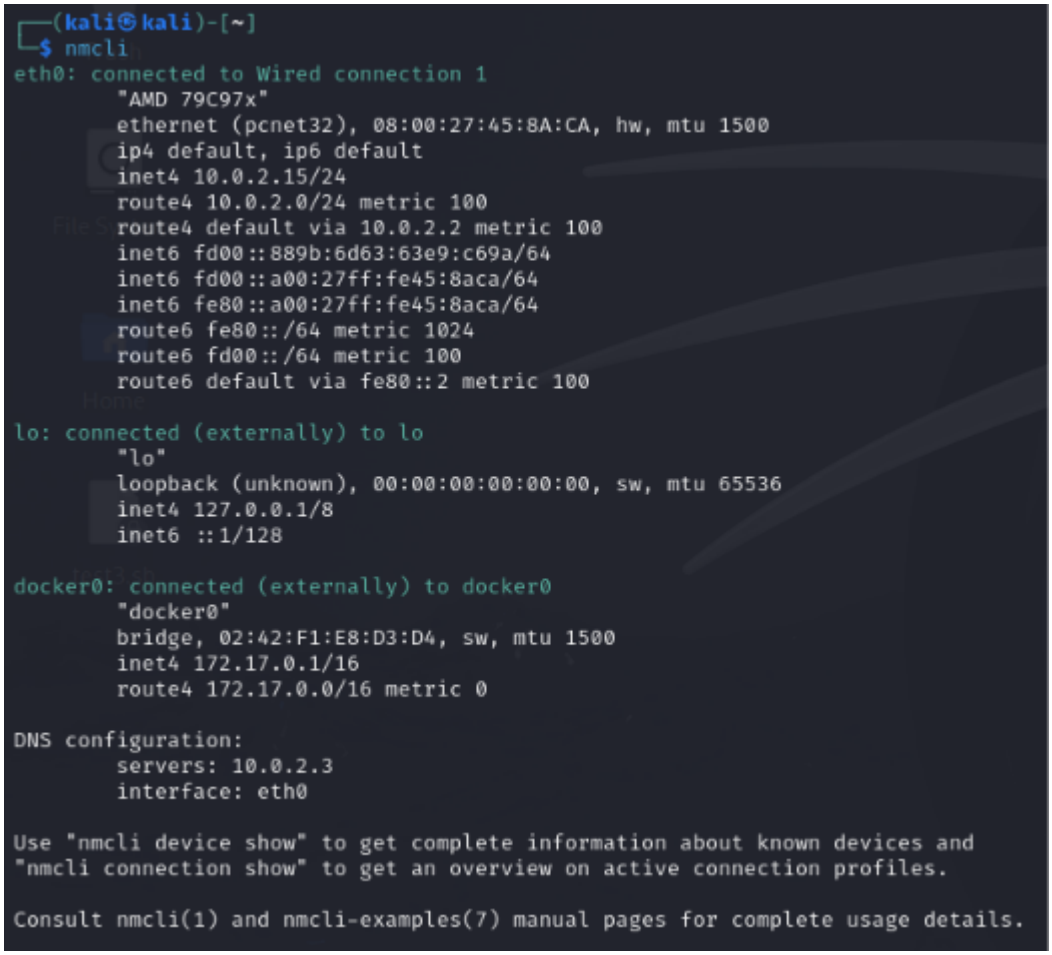


Figure 3: nmcli

Tool used to manage and inspect network connections through network manager.

Connectivity Testing

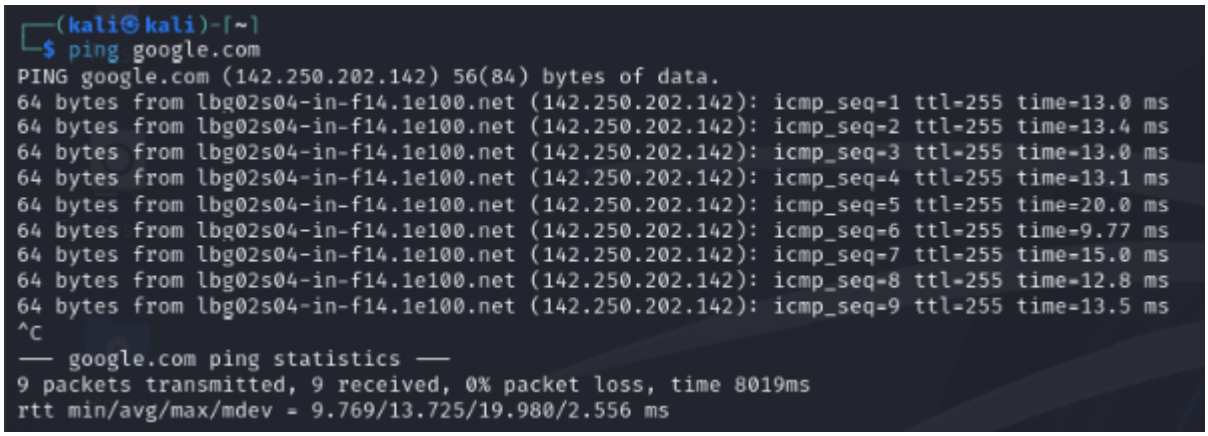


Figure 4: ping google.com

Tests network connectivity by sending IMCP echo request packets to the destination. It displays whether the destination responds, round trip response time, packet loss.

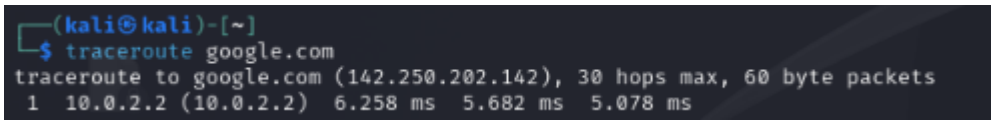


Figure 5: traceroute google.com

Displays the route that network packets take to reach the destination.

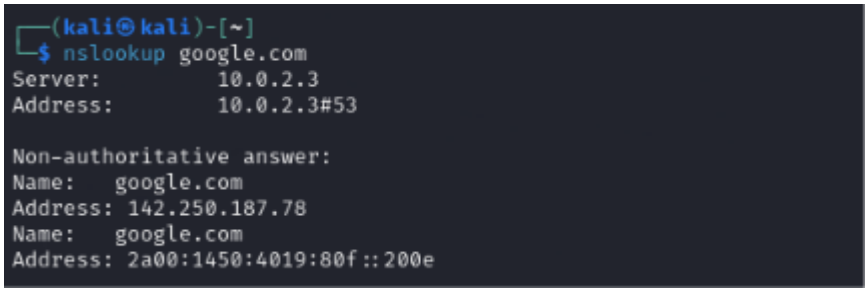


Figure 6: nslookup google.com

It queries DNS servers to retrieve information about the domain name.

Ports and Services

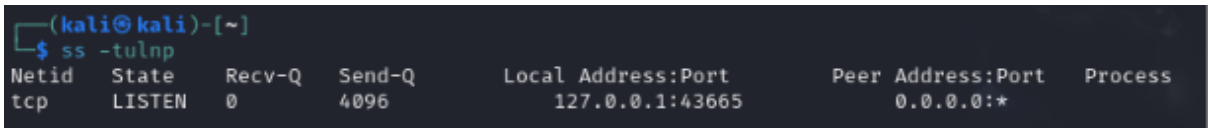


Figure 7: ss -tulnp

Displays information about network sockets and listening ports.

```
┌──(kali㉿kali)-[~]
└─$ netstat -anp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program nam
e
tcp        0      0 127.0.0.1:43665         0.0.0.0:*               LISTEN      -

udp        0      0 10.0.2.15:68            10.0.2.2:67             ESTABLISHED -

raw6       0      0 :::58                   :::*                    7           -

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   PID/Program name     Path
unix  3      [ ]         STREAM     CONNECTED     10317    1381/xfdesktop
unix  3      [ ]         STREAM     CONNECTED     13168    1636/gvfsd-trash     /run/user/1000/
gvfsd/socket-njC6xQ1m
unix  2      [ ]         DGRAM      CONNECTED     22308    -
unix  3      [ ]         STREAM     CONNECTED     9694     -                    @/tmp/.X11-unix
/X0
unix  3      [ ]         STREAM     CONNECTED     6500     -
unix  3      [ ]         STREAM     CONNECTED     9658     -                    @/tmp/.X11-unix
/X0
unix  3      [ ]         STREAM     CONNECTED     8294     -                    /run/systemd/jo
urnal/stdout
unix  3      [ ]         STREAM     CONNECTED     16737    1185/xfce4-session   @/tmp/.ICE-unix
/1185
unix  3      [ ]         STREAM     CONNECTED     16224    -                    @/tmp/.X11-unix
/X0
unix  3      [ ]         STREAM     CONNECTED     9662     1162/dbus-daemon     /run/user/1000/
bus
unix  3      [ ]         STREAM     CONNECTED     8956     1160/pipewire-pulse
unix  3      [ ]         STREAM     CONNECTED     22239    -
```

Figure 8: netstat -anp

It displays network connections, listening ports and associated processes.

Logs

```
┌──(kali㉿kali)-[~]
└─$ cat /var/log/syslog
2026-11-12T13:27:16.994066+04:00 kali systemd[1]: Listening on syslog.socket - Syslog Socket.
2026-11-12T13:27:16.995109+04:00 kali systemd[1]: Starting rsyslog.service - System Logging Ser
vice ...
2026-11-12T13:27:16.994347+04:00 kali rsyslogd: imuxsock: Acquired UNIX socket '/run/systemd/jo
urnal/syslog' (fd 3) from systemd.  [v8.2608.0]
2026-11-12T13:27:16.994437+04:00 kali rsyslogd: [origin software="rsyslogd" swVersion="8.2608.0
" x-pid="3359" x-info="https://www.rsyslog.com"] start
2026-11-12T13:27:16.995141+04:00 kali systemd[1]: Started rsyslog.service - System Logging Serv
ice.
2026-11-12T13:27:16.994264+04:00 kali kernel: Linux version 6.11.2-amd64 (devel@kali.org) (x86_
64-linux-gnu-gcc-14 (Debian 14.2.0-6) 14.2.0, GNU ld (GNU Binutils for Debian) 2.43.1) #1 SMP P
REEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15)
2026-11-12T13:27:16.995532+04:00 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.11.2-amd
64 root=UUID=15192a85-7d0e-49ee-b334-b4e46d613fe2 ro quiet splash
2026-11-12T13:27:16.995534+04:00 kali kernel: BIOS-provided physical RAM map:
2026-11-12T13:27:16.995534+04:00 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009
fbff] usable
```

Figure 9: cat /var/log/syslog

Displays the contents of the system log file. Helps investigate system activity and troubleshoot errors.

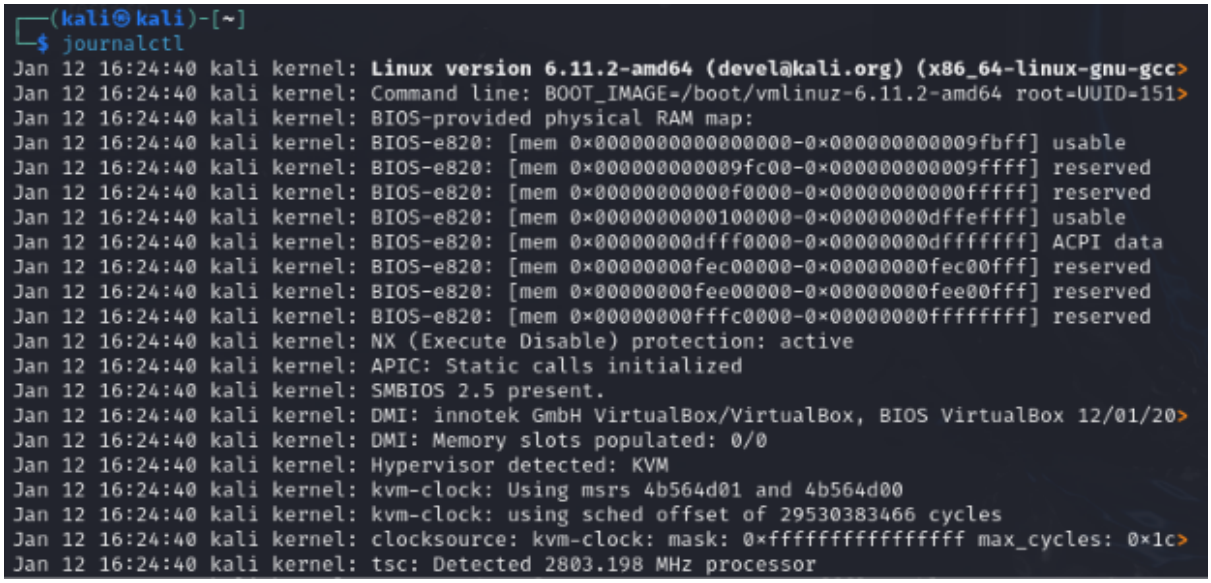


Figure 10: journalctl

It displays system logs collected by systemd's journal.

Hands on Task

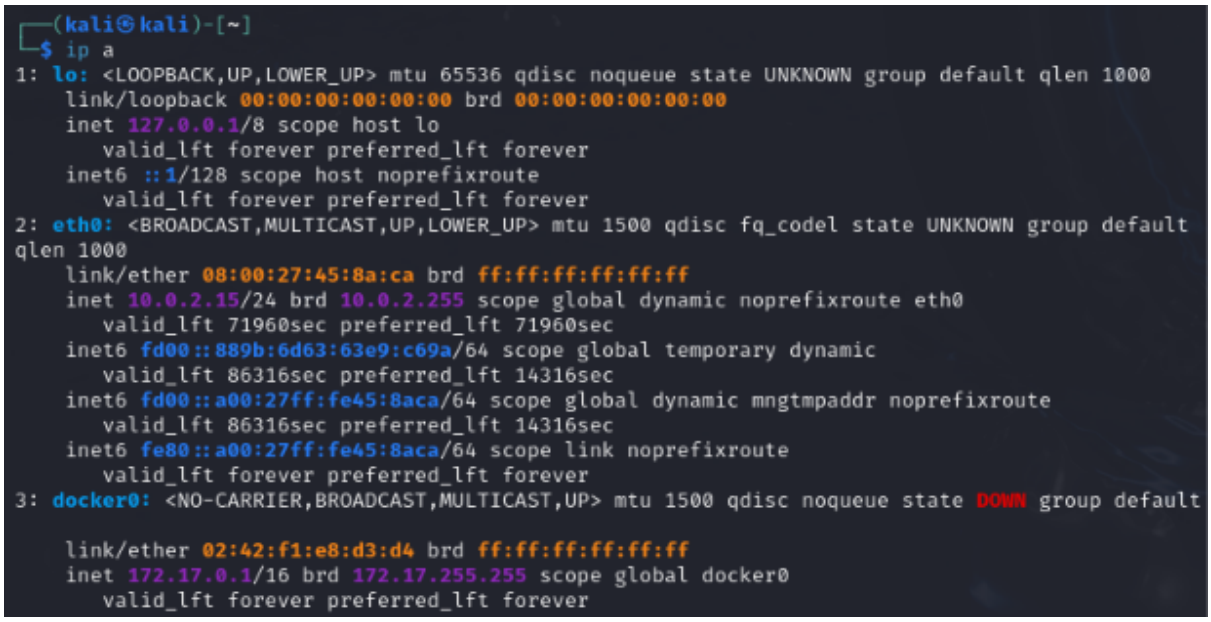


Figure 11: Identifying IP address.

My IP address is 10.0.2.15.



Figure 12: Identifying default gateway

My default gateway is 10.0.2.2.

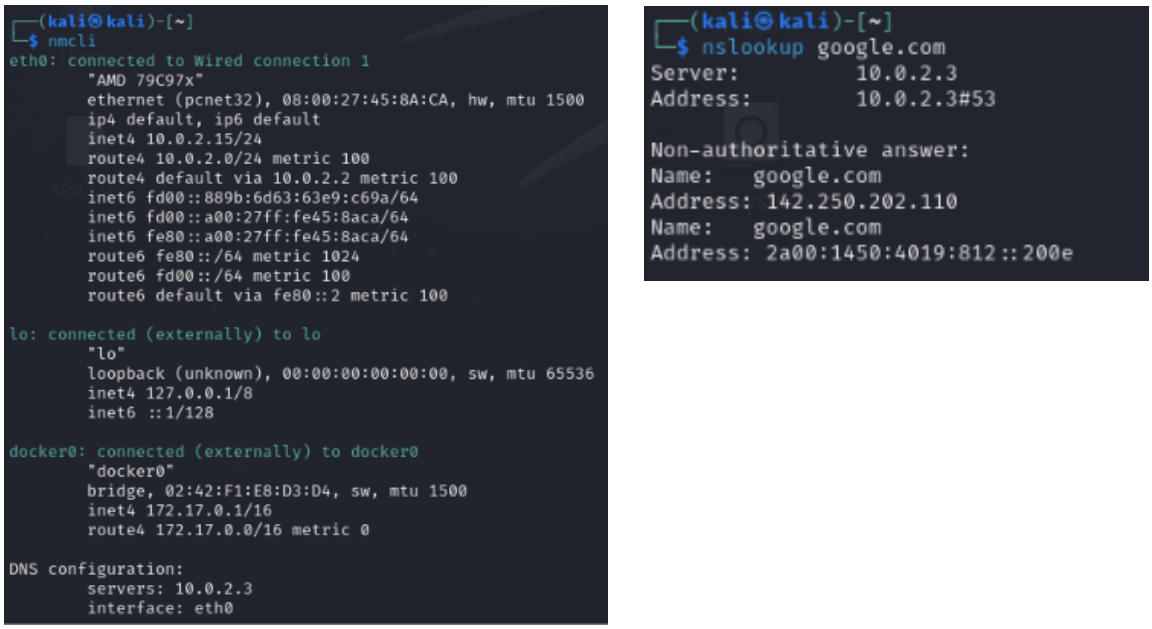


Figure 13: Identifying DNS server
My DNS server is 10.0.2.3.

TCP (transmission control protocol) is a connection oriented protocol which provides reliable data transmission. It establishes a connection before sending data using a three way handshake. It uses acknowledgments and retransmission (resend missing packets) to ensure that data packets have reached the destination correctly and in order. Example: https, email, SSH.
Whereas UDP (user datagram protocol) is a connectionless protocol that sends data packets without establishing connection and guaranteeing delivery or packet order. It doesn't retransmit missing packets. It sends data packets without waiting for acknowledgments. Its useful for applications that require fast communication and low latency, example: Gaming, Streaming and calls etc.